



Global Ocean Pollution Index

A Data Science Project — Full Project Report

oceanpollutionindex.com

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Contents

Contents	2
1. Project Overview	3
At a Glance	3
2. The Problem It Solves	3
3. The Data	3
4. How the Index Is Calculated	4
5. Forecasting to 2100	5
6. The Website	5
Landing Page	5
Interactive Dashboard	5
About Page	6
7. Key Findings	6
8. Technical Stack	6
The Pipeline	6
9. Limitations & Honesty	6
10. Why This Project Matters	7

1. Project Overview

The Global Ocean Pollution Index is an end-to-end data science project that measures, scores, and forecasts pollution across 35 of the world's major water bodies — oceans, seas, and large lakes.

It takes pollution data that is scattered across more than a dozen government and scientific agencies — in formats most people never open — and brings it together into a single, comparable score from 0 to 100 for each water body. Every region is then forecast forward to the year 2100.

The result is a live, interactive website where anyone can explore a world map, click any ocean or sea to see what is driving its pollution, and drag a slider to see how the future changes depending on how much action the world takes.

What makes it more than a chart: the project covers the full data science lifecycle — sourcing raw data, cleaning and merging it, building a scoring model, forecasting, and deploying a polished public product.

At a Glance

Metric	Detail
Water bodies	35 oceans, seas, and large lakes
Pollution factors	12 weighted factors per region
Data sources	13+ public datasets (NOAA, UN, IPCC, Copernicus, OECD, FAO, OHI)
Forecast horizon	2026 → 2100
Live site	oceanpollutionindex.com

2. The Problem It Solves

Pollution data for the world's oceans already exists — but it is fragmented. Microplastic readings sit with NOAA, acidification data with Copernicus, population figures with the UN, climate projections with the IPCC, and so on. Each uses different formats (CSV, NetCDF, GeoJSON), different geographic scales, and different units.

Because of this fragmentation, there was no single, comparable answer to a simple question: which water bodies are most polluted, and where are they headed?

This project answers that question by doing the hard integration work — turning many incompatible sources into one clean, comparable index.

3. The Data

Twelve pollution factors feed the index, drawn from the following public datasets. Every source is free and openly licensed; missing regional values were filled using published peer-reviewed literature, flagged in the code.

Factor	Source	What it captures
Microplastics	NOAA NCEI	Plastic particles in the water
River plastic	Our World in Data	Plastic flowing in from rivers
Dissolved oxygen	NOAA WOA	Dead zones / oxygen loss
Port pressure	World Port Index	Industrial & shipping activity
Coastal population	Copernicus	People within 10km of shore
Oil spills	NOAA IncidentNews	Spill frequency & volume
Ocean pH	Copernicus Marine	Acidification from CO ₂
Sea temperature	NOAA WOA	Thermal stress
Wastewater	FAO AQUASTAT	Sewage & farm runoff
Clean water	Ocean Health Index	Independent contamination score
Biodiversity	Ocean Health Index	Marine species & habitat health
Plastic mismanagement	OECD	Share of badly disposed plastic

Two further datasets — the UN's World Population Prospects and the IPCC's CMIP6 climate projections — are used not for the present-day score but for forecasting (see Section 5).

4. How the Index Is Calculated

Each of the twelve factors is normalised to a 0–100 scale using min-max scaling across all 35 regions. The factors are then combined using a weighted average to produce a single pollution score per water body.

The weights reflect how directly each factor measures pollution. Microplastics and river plastic carry the most weight because they are the most direct measures; supporting factors like biodiversity and plastic mismanagement carry less.

Factor	Weight
Microplastic concentration	18%
River plastic input	15%
Dissolved oxygen depletion	10%
Port & shipping pressure	9%
Coastal population	8%
Oil spill pressure	7%
Ocean pH (acidification)	7%
Sea surface temperature	7%
Wastewater & runoff	5%

Clean water (OHI)	5%
Biodiversity (OHI)	5%
Plastic mismanagement	4%

Scores are then banded into grades — Minimal, Low, Moderate, High, Critical, Severe, Extreme, and Catastrophic — which drive the colour scale on the map.

5. Forecasting to 2100

A naive forecast would apply one growth rate to every region. This project does something more sophisticated: each water body is forecast using its own growth rate, blended from three independent real-world data sources.

Source	Weight	What it adds
Microplastic trend	55%	How fast plastic has actually risen in each region over 50 years of NOAA data, via regression
Population growth	25%	The UN's real projection for each region — some grow, some shrink
Ocean warming	20%	The IPCC's temperature projection for that ocean region

This means regions with rising plastic, growing populations, and rapid warming climb steeply — while regions like the Mediterranean, where the plastic trend is falling and the population is shrinking, are actually projected to improve. The forecast tells a different story for every region, grounded in real data.

6. The Website

The project is delivered as a three-page website, all built with vanilla HTML, CSS, and JavaScript (no framework).

Landing Page

Introduces the project, shows headline statistics (35 water bodies, 12 factors, 2100 horizon), and presents the most striking findings before directing visitors to the dashboard.

Interactive Dashboard

The heart of the project. It includes:

- A dark-mode world map with colour-coded circles sized and shaded by pollution severity
- A year toggle to switch between 2026, 2050, and 2100
- A policy scenario slider — drag it to see the future under different levels of action, from 'worse' through 'now' to 'full action'
- A clickable detail panel showing every factor for a chosen region
- A forecast chart for the five most polluted regions

- A league table ranking all 35 regions across 2026, 2050, and 2100

About Page

Explains the methodology in full — the data sources, the factor weights, the three-source forecast, and an honest list of the project's limitations.

7. Key Findings

- The North Pacific ranks highest today — it has the lowest ocean pH of any basin and by far the highest concentration of recorded oil spill incidents.
- The Indian Ocean carries the heaviest human pressure: 208 million people live within 10km of its shores.
- The Great Lakes (Erie, Ontario, Michigan) rank above most oceans, driven by extreme microplastic densities — pollution from plastic, not ships or oil.
- The Arctic scores low today but is warming faster than any region on the map, making it a future hotspot.
- Not every region worsens: the Mediterranean is projected to improve by 2100, thanks to a falling plastic trend and shrinking coastal population.

8. Technical Stack

Layer	Tools used
Data processing	Python — pandas, NumPy, xarray, scipy, netCDF4
Data formats	CSV, NetCDF, GeoJSON, Shapefile
Visualisation	Leaflet.js (map), Chart.js (charts), vanilla HTML/CSS/JS
Version control	Git & GitHub
Hosting	Firebase Hosting, custom domain via GoDaddy

The Pipeline

The project runs as a sequence of Python scripts, each feeding the next:

- step1 — loads and audits every raw dataset
- step2 — cleans, merges all 12 sources, and calculates the index
- step2b — calculates each region's historical growth rate via regression
- step2c — blends the three forecast sources into a per-region forecast
- step3 — generates the interactive dashboard from the final data

9. Limitations & Honesty

A credible project is open about its constraints. These are stated openly on the site and were deliberate decisions:

- The factor weights are chosen by reasoning about relevance, not derived statistically — a standard but subjective approach for composite indices.
- Microplastic units vary across studies (items/m³, items/kg, items/km²), so normalisation across measurement methods is approximate.
- Oil spill data is strongest for US coastal waters, so some regions are likely under-represented.
- Forecasts assume current trends continue — they are trajectories, not predictions.
- When expanding the index, only water bodies with genuine, comparable water-column data were added. Regions without solid data were deliberately excluded rather than filled with placeholder values.

That last point is worth emphasising: the harder decision was often what to leave out, not what to include. A smaller, internally consistent dataset is more defensible than a larger one stitched from incompatible measurements.

10. Why This Project Matters

This is a genuine end-to-end data science project. It moves from scattered raw data all the way to a live, interactive public product — covering data engineering, modelling, forecasting, visualisation, and deployment.

The hardest and most valuable part was the data engineering: taking more than a dozen messy, incompatible sources and turning them into one clean, comparable index. That integration work — not the code alone — is what data teams most need and most struggle to find.

Every decision in the project can be explained and defended, from the choice of weights to the regions left out. That combination of technical depth and honest judgement is what the project demonstrates.